

Course: B. Tech in Civil Engineering

Sem: IV

Subject Name: Surveying-II (BTCVC402)

Max Marks: 60

Date: 28/11/2019

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

Marks

Q.1 Solve the following.

- A) i) Explain instruments used in Tacheometry. 06
 ii) Explain characteristics of Tacheometry.
 iii) Explain importance of EDM
- B) Following observations made using a tacheometry fitted with an anallatic lens, the multiplying constant being 100. 06

Inst. station	Height of Inst.	Staff station	WCB	Vertical angle	Hair readings	Remarks
O	1.550	A	30°30'	4°30'	1.155, 1.755, 2.355	RL of O =
		B	75°30'	10°15'	1.250, 2.000, 2.750	150.000

Calculate the distance AB, the RLs of A and B. Find also the gradient of the line AB.

Q.2 Solve the following.

- A) Explain measurement of horizontal distance by Subtense Bar and Procedure of observation with neat figure. 06
- B) Explain Triangulation systems/figures with neat diagrams. 06

Q.3 Solve the following.

- A) Define: 06
- i) The Zenith iii) The Latitude v) The Altitude
- ii) The Nadir iv) The Longitude vi) The Declination

- B) Explain Spherical triangle with their properties with formulae in Spherical Trigonometry. 06

Q.4 Solve the following.

- A) Define Transition curve. Explain requirements of Ideal transition curve. Explain objectives of providing transition curves. **06**
- B) Explain in detail: i) Degree of Curve ii) Relation between radius and degree of curve iii) Superelevations. **06**

Q.5 Solve the following.

- A) i) A vertical photograph was taken at an altitude of 1200 m above mean sea level. Determine the scale of the photograph for terrain lying at elevations of 80 m and 300 m if the focal length of the camera is 15 cm. **06**
- ii) A camera having focal length of 20 cm is used to take a vertical photograph to a terrain having an average elevation of 1500 m. What is the height above sea level at which an air-craft must fly in order to get the scale of 1:8000?
- B) a) Explain: **06**
- i) Mosaic and its type.
- ii) Stereoscopy and Photo interpretation.
- b) The scale of an aerial photograph 20cm x 20cm is 1cm=100m. Determine the number of photographs required to cover an area of 8km x 12.5km, if the longitudinal overlap is 60% and side overlap is 30%.

Q.6 Solve any one of the following.

- A) Write uses and applications of i) G.P.S. and ii) G.I.S. in Civil Engineering. **06**
- B) Explain the Idealized Remote sensing system and their stages with neat sketch **06**

***** Paper End *****